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Using Artificial Neural Networks to Forecasting Carbon Dioxide Emissions in Iraq

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***Abstract*-This paper explores the application of ANN (artificial neural networks) to forecast CO_2 emissions in Iraq until 2028. ANNs are able to model non-linear dynamics of time series data which eventually leads to accurate forecasts without any statistical assumption about the features of a dataset. The authors developed a simple single-input feedforward ANN model with the yearly CO_2 emission data from 1991 to 2023 as the input to project the future emissions using the year. The Levenberg-Marquardt algorithm was used for network training. The model performed well on the training, validation, and testing datasets with minimal error rates and R-squared values of 1, which implied that the regression demonstrated a good fit between targets and outputs. The performance of ANNs in forecasting was evaluated. The mean squared error (MSE=0.1325) and root mean squared error (RMSE = 0.3641) values were low, highly predictive of small forecasting errors. R^2 is quite high (0.946), indicating the model could explain as much as 94.6% of the variances in the actual data. The mean absolute percentage error equaled 8.01%, which signifies a good forecast with less than 10% error. The forecast of 2028 shows per capita emissions reaching 3.649 tons, which may be affected by population growth, economic development, or infrastructure changes that will be put into place. Despite renewables, efficiency, and emissions control or policies the growth curve can be replaced. This model serves as a data-driven instrument for future Iraqi CO_2 emissions forecasting to develop climate change mitigation policies which are not time series statistical assumptions. It could also be extended to other greenhouse gases and countries, which is possible. This paper shows that ANNs can predict emissions that are accurate and reliable for decision-making which helps to reduce the country's carbon footprint and climate change.**

***Keywords*—Prediction; backpropagation algorithm; time series analysis; environmental forecasting in Iraq; climate change.**

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I. INTRODUCTION

One of the main objectives of statistical sciences is to predict the future behavior of the phenomenon in question. Where we find daily, through the media, daily interest in weather forecasts and various economic variables. Time series analysis has paid much attention to forecasting and several methods have been proposed for the purpose of making a prediction [1]. Artificial Neural Networks (ANNs) are among the modern methods that have received noticeable attention, especially in the field of computer science, as the basic idea of this method is to create an information model that simulates a neurobiological system, such as the brain, and information processing. The main key to this model is to build a new structure for the information processing system that links and organizes many interrelated processing elements (neurons) that work in concert to solve the problem under study [2].

Artificial Neural Networks (ANNs) are designed and structured for certain purposes, like model discrimination and perception or data classification through the learning process. ANNs learn similarly like humans do, through examples and training. Furthermore, the fundamental principle of neural networks is that learning occurs through conditioning connections between neurons in biological systems [3]. Neural networks have recently piqued interest in the statistical community for use in statistical applications. It is thought that using the neural network approach could be beneficial in resolving the prognosis problem we are addressing in this study because it does not require assumptions about the nature of the time series, whether it is linear, natural, or phased [4]. Then, Alardhi et al. [5] add to this idea, in their research, the artificial neural network (ANN) method that was employed to predict the concentration of H_2S and CO_2 in the sweetening

process. The information includes the H_2S and CO_2 levels of the feed gas, the temperature, pressure, and flow rate of the device. The designed ANN model exhibited accurate modelling of the process at hand, even for a large variety of parameter variations [5].

In the same year, various numbers of neurons were incorporated into the models' structures. It is demonstrated that utilizing MLP ANN enables the accurate prediction of the CO_2 emissions of these nations. Furthermore, it is determined that networks with radial basis and tansig transfer functions have the lowest mean squared error (MSE) and maximum accuracy when employing logsig transfer function. The networks with logsig, radial basis, and tansig transfer functions have R-squared values of 0.9998, 0.9997, and 0.9996, in that order. Lastly, a comparison between the suggested model and related research that looked at five nations in the same region shows improved accuracy in terms of MSE [6].

Shabri [7] handles the prediction of CO_2 emissions was modelled using three distinct models: the gray model GM(1,N), the artificial neural network (ANN), and the GMDH models. The comparison showed that the Lasso-GMDH model had the best overall performance for estimating Malaysia's yearly CO_2 emissions. The study uses data from the European Centre for Medium-Range Weather Forecast (ECMWF) [7] and investigate the chemical behavior of greenhouse gases (CH4, CO_2) and their effects on climate change in Iraq from 2003 to 2017. The study uses data from the European Centre for Medium-Range Weather Forecast (ECMWF) to examine the chemical behavior of greenhouse gases (CH4, CO_2) and their effects on climate change in Iraq from 2003 to 2017 [8], [9]. For the global climate and weather, the researcher Copernicus Atmosphere Monitoring Service (CAMS) ECMWF in 2022, the analysis was conducted. The findings also showed that in June, the atmosphere's concentration of CO_2 was at a high level, with the maximum limit recorded throughout all of Iraq's regions being about 396.6 ppm [10]. From what researchers in Iraq and some other countries have presented in recent years, it has become clear that carbon dioxide emissions have not been analyzed, and its impact was not studied and predicted using neural networks. The current study presented a procedure for analyzing carbon dioxide emissions and predicting it in the future to see the impact of gas emissions alone in the atmosphere of Iraq.

II. Materials and Methods

A. Artificial Neural Networks (ANN)

A neural network is a set of algorithms and data structures that simulates the operation of the human brain in the field 8of information technology. A neural network has many processors that operate in parallel. The process of "training," or feeding large amounts of data and rules about the relationships between those data, establishes the neural network's initial state. Then, a program instructs it on how to react to external stimuli, which can come from the user's input or can be initiated by the program itself by carrying out a specific task [11].

Neural networks are based on simple mathematical models that have certain performance characteristics in a manner that simulates biological neural networks (nervous system) such as the brain [6]. Fig. 1 shows the shape of the neural network.

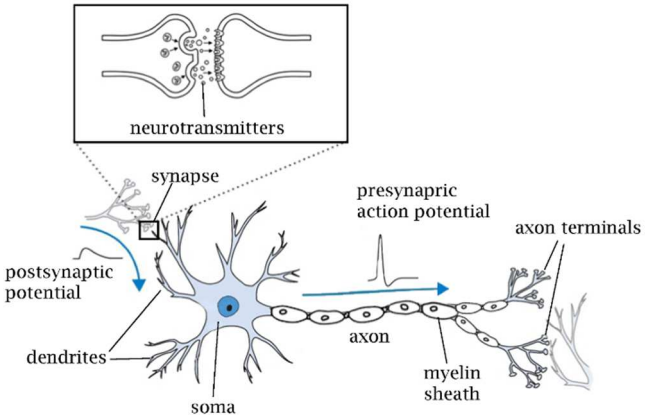


Fig. 1 Schematic of a Biological Networks [11], [12]

The architecture of an information processing system that links and organizes numerous internally linked processing units (neurons) that collaborate to solve specific problems is the key to the artificial neural network model. Signals between nodes (neurons) pass through connecting lines, and each line is attached to a specific weight, and the signals are multiplied [7]. These weights are sent into the neuron, or node, and the nodes or neurons then aggregate the weighted inputs. The activation function, a non-linear function with a threshold, processes each node's outputs after that [13]. Fig. 2 shows a simple neural network.

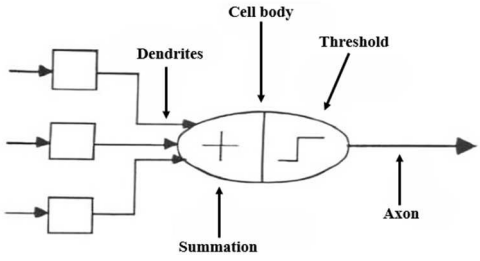


Fig. 2 Simple artificial neural network [3]

As is well known, the neural network is a group of neurons and organized weights, and the neurons are organized into layers, and each neuron in each layer has a weight associated with each neuron in the next layer [14] Artificial neural networks learn by certain means to change the strength of the internal linkage, and this is called "weight conditioning" and it occurs during a process called the "training phase". The external input set is called the “training set of the network” or (training sample), and the single input model is called the “training vector” or (observations), in addition to that the network can receive via feedback, or it can use the feedback with additional information to determine the size of change in weights.

The network gains knowledge by being given a series of carefully chosen examples, which will accelerate the network's learning process. Depending on the training class that is given to the neural network, there are two sections to the procedures for teaching it [15], they are as follows:

1) Supervised Learning: In this method, the networks need a trainer to teach the network the desired output, and the

training class is a pair of vectors, the input vector which is the values entered into the network, and the output vector which is the values that the network should output, including the Backpropagation Network.

2) Unsupervised learning: In this method the network adapts easily in response to the inputs and learns to choose the hierarchy from the inputs only and the training class is the input vector only [16].

B. Multilayer Perceptions Neural Network Models

Neural network models called MLPs serve as universal approximators, able to approximate any continuous function. For instance, they could be used as SEE models. The neurons that comprise MLPs are known as perceptions. Thus, the overall structure of perceptron and the fundamental structure of MLPs will be discussed first. Each of the n features (x = x1, x2, ..., xn) that are fed into a perceptron, as seen in Fig. 3, is given a weight. Numerical input features are required. Therefore, nonnumeric input attributes need to be converted into numeric ones in order to use a perceptron [17]. For example, p potential values in a categorical feature can be converted into p input features that show whether these values are present or not.

After receiving the input features, an input function u calculates the input features' weighted sum:

$$u(\mathbf{x}) = \sum_{i=1}^{n} w_i x_i \tag{1}$$

After obtaining the result of this computation, an activation function f generates the output of the perceptron. Step functions are used as the original perceptron's activation function [18]

$$y = f(u(\mathbf{x})) = \begin{cases} 1, & \text{if } u(\mathbf{x}) > \theta \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

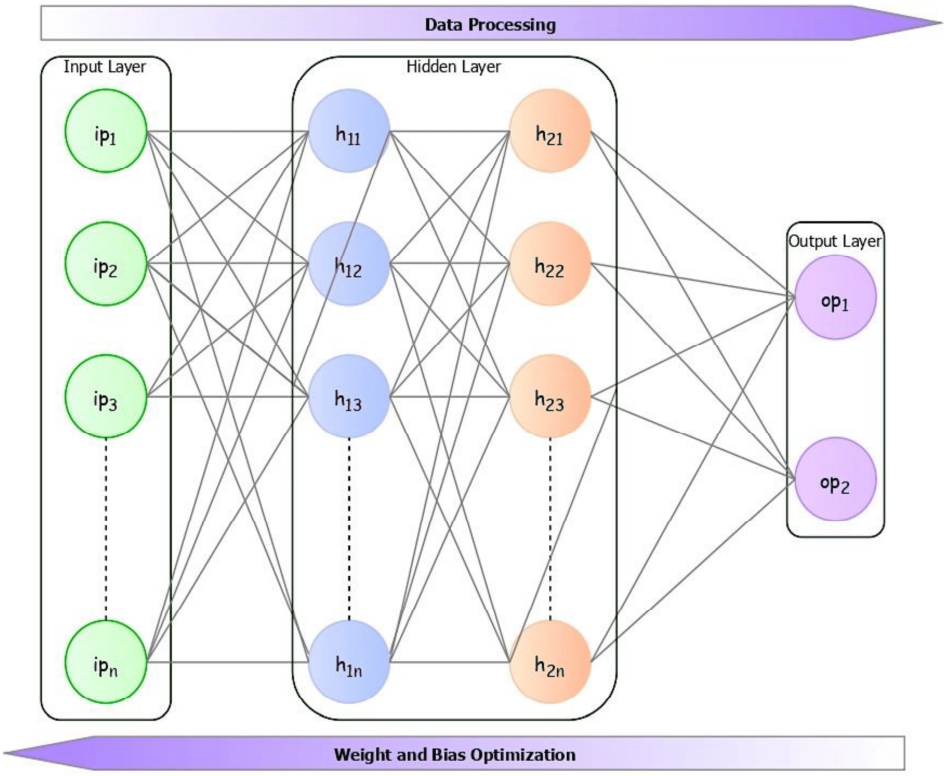


Fig. 3 A perceptron's scheme with n input features [19], [20]

where θ is a threshold parameter. An example of step function with $\theta = 0$ is shown in Fig. 4. Thus, we can see that the perceptron determines whether $w_1x_1 + w_2x_2 + \cdots + w_nx_n - \theta > 0$ is true or false. The equation $w_1x_1 + w_2x_2 + \cdots + w_nx_n - \theta = 0$ is the equation of a hyperplane. T For any input point above the hyperplane, the perceptron outputs 1, while for any input point on or below the hyperplane, it outputs 0. The perceptron is referred to be a linear classifier because of this; that is, it performs well with data that can be divided linearly. The process of perceptron learning involves varying the weights in order to identify a hyperplane that effectively divides the training set. We recommend Gurney [21] for additional details regarding the learning algorithm of the perceptron.

Rather than just approximating linear functions, MLPs may approximate any continuous function. They accomplish this by merging several neurons, each of which is arranged into at least three layers (see Fig. 5):

- A single input layer that merely transfers the input features to the top-level hidden layer.
- A concealed layer or levels of perceptions. The features dispersed by the input layer are received as inputs by the first hidden layer. Each perceptron's output from the preceding layer is sent into the other hidden levels.
- One output layer of perceptions, which takes each perceptron's output from the final hidden layer as an input [22].

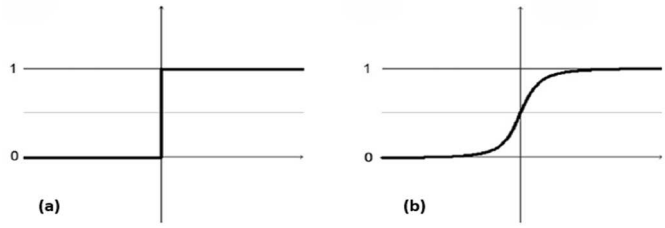


Fig. 4 Function for activation; (a) step function and (b) sigmoid function [19]

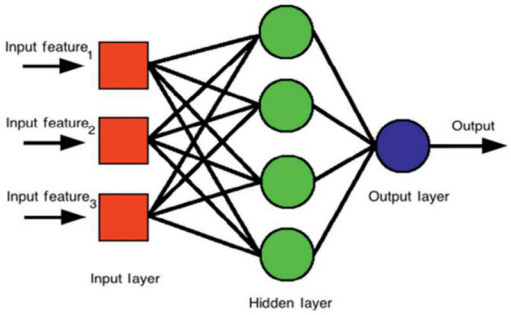


Fig. 5 Diagram of a three-layer MLP consisting of one output, four hidden neurons, and three input characteristics [23], [24]

C. The Network Algorithm Includes Five Steps to Forecasting:

The process of forecasting data with a network algorithm involves multiple steps. These actions usually consist of:

1) Data Preparation: $X = \{x_1, x_2, \dots, x_T\}$ where x_t represent the observation at time t and split data into training set X_{train} and testing set X_{test} [25].

2) Model Architecture: Feedforward neural network with L layers where input layers have n_0 nodes (resembling the input window size), and output layer n_L [26].

3) Model Initialization: Weights $W^{(l)}$ and biases $b^{(l)}$ of the $l-th$ layers are initialized with small random values.

4) Forward Propagation: For the input layer: $a^{(0)} = x_t$ so for hidden layers ($l = 1,2, \dots, L-1$):

$$z^{(l)} = W^{(l)} a^{(l-1)} + b^{(l)}$$
$$a^{(l)} = f\left(z^{(l)}\right) \quad (3)$$

where f is the activation function [27]. And for the output layer ($l = L$):

$$z^{(L)} = W^{(L)} a^{(L-1)} + b^{(L)}$$
$$\hat{x}_{t+1} = a^{(L)} = z^{(L)} \quad (4)$$

5) Backpropagation: The Loss function J measures the error between the predicted output $\hat{x}_{t+1}$ and the actual target x_{t+1}.

A common choice means squared error (MSE) [27], [28].

$$J = \frac{1}{2}(x_{t+1} - \hat{x}_{t+1})^2 \quad (5)$$

Then, computing gradients of the loss function J w.r.t. weights and biases using the chain rule:

$$\frac{\partial J}{\partial W^{(l)}} = \frac{\partial J}{\partial z^{(l)}} \frac{\partial z^{(l)}}{\partial W^{(l)}}$$
$$\frac{\partial J}{\partial b^{(l)}} = \frac{\partial J}{\partial z^{(l)}} \frac{\partial z^{(l)}}{\partial b^{(l)}} \quad (6)$$

where Error is backpropagated from output layer to input layer, updating gradients for each layer [27], [28].

6) Weight Update: Update weights and biases using an optimization algorithm like Levenberg-Marquardt:

$$W^{(l)} = W^{(l)} - \eta \frac{\partial J}{\partial W^{(l)}}$$
$$b^{(l)} = b^{(l)} - \eta \frac{\partial J}{\partial b^{(l)}} \quad (7)$$

where η is the learning rate [29]. After this step, steps 3-6 are repeated for multiple epochs over the training data. Then, forecasting errors (e.g., MSE, MAE) on testing set X_{test} to assess model's performance is computed.

7) Forecasting: The trained neural network has been used to make forecasts $\hat{x}_{t+1}$ on new and unseen time series data by performing forward propagation with the learned weights and biases [30].

III. Results and Discussion

It is represented in the emissions of carbon dioxide (CO_2), as it is the most polluting gas, especially when using fossil energy sources to produce electricity. Iraq faces great challenges due to the high rates of carbon dioxide gas generated, especially from land transportation, as well as the presence of factories and oil and gas refineries. Therefore, it is necessary to know the behavior of this phenomenon in the future through the method of forecasting using artificial neural networks, as it is more accurate than the traditional methods of forecasting.

A. Description of the Time Series (CO_2)

One of the first steps to follow to describe or analyze a time series is to draw it with what is known as the Time Series Plot, where it is possible through the drawing to see the nature of the fluctuation in it and note whether it includes a general trend or not? Whereas, knowing the general trend is very useful in ascertaining the stationarity of the time series, as well as its close relationship to forecasting.

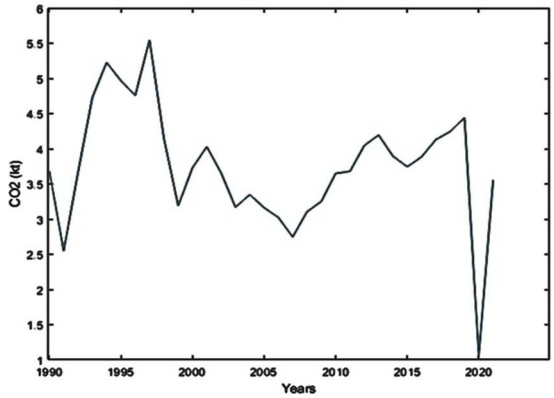


Fig. 6 Carbon dioxide emissions evolution in Iraq (kilotons)

Fig. 6 displays the graph of the time series under consideration. It is observed that the series' fluctuation starts off slowly declining before gradually rising in the shape of a second-degree parabola, indicating that the series is not phasing. However, we also see a distinct dispersion around the overall oscillation pattern, indicating the presence of significant and distinct random effects in this series.

B. Application of Backpropagation algorithm

A computer program designed in the MATLAB language was used to obtain predictions and calculate statistical parameters. A backpropagation neural network was built for the original observations, 2 hidden layers.
The results of this algorithm will be presented in three steps, as follows:

1) Step 1: Setting the initial values, stopping criteria, and target values for the training processes. In this step, the initial epoch, the time before training starts, and the performance of the network using the initial mean square error criterion, the gradient coefficient, and the mu coefficient will be determined initially. Then, after the training process, we will get the stop values, as shown in Table I.

TABLE I
SETTING INITIAL TRAINING PARAMETERS

Unit	Initial Value	Stopped Value	Target Value
Epoch	0	1000	1000
Elapsed	-	00:02:14	-
Performance (MSE)	5.5	1.59e-07	0
Gradient	11.8	5.24e-07	1e+17
Mu	0.001	1e-07	1e+10
Validation Checks	0	0	50

From Table I, we can see that the initial epoch value before the training process is 0. After the training process, the algorithm stopped at epoch 1000, and the planned target number of epochs is 1000. The time taken to train the network is 10 seconds. In terms of the performance of the network, i.e., the mean square error criterion, the initial value before the training process is 3. 67. After the training process, it becomes 1.4e-10 at epoch 1000. The performance value that we aim to achieve after training is 0. In terms of the gradient coefficient, the value before the network training is 9.98. After the training process, it becomes close to zero, i.e 1e-06, which is very close to the planned value of 1e-07. In terms of the Mu coefficient, the initial value is 0.001. The value after the training process is almost equal to 1e-08. In terms of validation checks, the value is 50 after the training process, while it was 0 before the training process. The training dataset comprises 21 observations, while the testing and validation datasets each consist of 5 observations.

2) Step 2: Defining the training algorithm. The network was trained using the Levenberg-Marquardt learning algorithm, which consumes more memory but takes less time for calculations. The number of data samples taken for analysis, the mean square error (MSE) value was 1.59e-07, and the regression (R) value obtained after training are displayed. MSE is the average square difference between the outputs and the targets. Lower MSE values are better. Regression values are used to measure the relationship between the outputs and the targets. An R value close to 1 indicates a close relationship, while a value of 0 indicates a random relationship. Training sample results vary several times due to different initial conditions and sampling methods. The sample training process is performed over 1000 epochs, so the best network was achieved at 1000 iterations, 4 neurons, and a learning rate of 0.01, as shown in Table II.

TABLE III
TRAINING ALGORITHMS

Data division	Random	Dividerand
Training	Levenberg-Marquardt	trainlm
Performance	Mean Squared Error	mse
Calculations	MEX	

From Table II, we can see that the training data was divided randomly using the dividerand command, which means that the training samples will be selected randomly. The training algorithm is Levenberg-Marquardt, which is specified using the trainlm command. The performance criterion is the mean square error.

3) Step 3: Defining the neural network structure: From Fig. 7, as below which represents the architectural structure of the neural network, shows that the number of inputs is 1, represented by the feature carbon dioxide emissions value, with 4 hidden layers and a tansig activation function. For the output layer, the pureline activation function, i.e., the linear function, was used. The number of outputs was 1, which represents the predicted values.

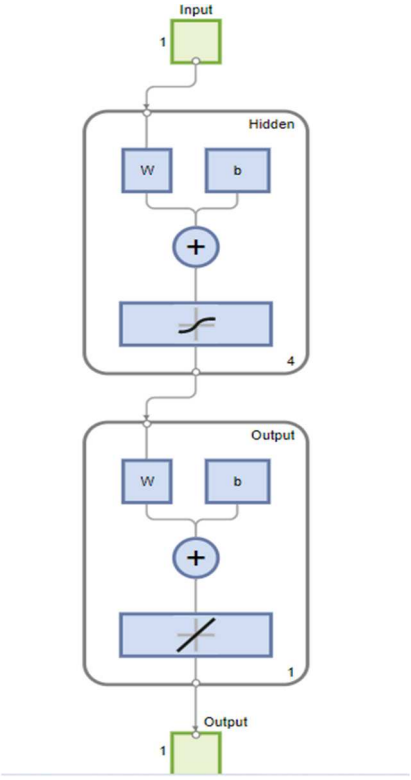


Fig. 7 Represents the architectural structure of the neural network Based on MATLAB 2022a

4) Step 4: Performance of the neural network. In this step, we will show the performance of the artificial neural network through the epoch values and the mean square error values, as shown in Fig. 8, which represents the network performance.

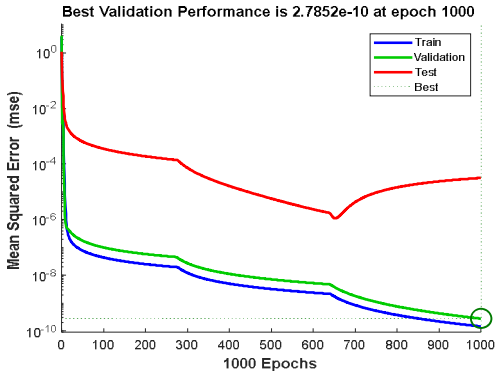


Fig. 8 Performance of the neural network

From Fig. 8, we can see that the best performance of the neural network is at epoch 13, with a mean square error value of 2.7852e-10, which is represented by the dotted black line. The green line shows the mean square error of the validation process, the red line shows the mean square error of the testing process, and the blue line shows the mean square error of the training process.

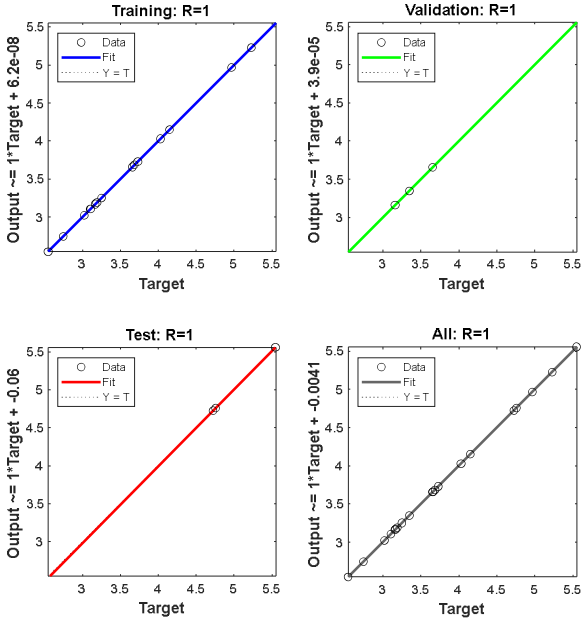


Fig. 9 The coefficient of determination R

Also, we noted from Fig. 9 the evaluation of the regression of target values to outputs is done for the processes of training, validation, and testing, as well as for the network as a whole. This is achieved through the coefficient of determination, which represents the interpretive strength of the artificial neural network model. Its values range between (0, 1), whereas it approaches one, it indicates the model's strength, and as it moves away from zero, it signifies the model's weakness. This is illustrated in Fig. 9, which represents the coefficient of determination for the network.

From Fig. 9, we can observe that the coefficient of determination R for the training process is equal to (1). This signifies that the target explains 100% of the variations occurring in the network's output. Similarly, the coefficient of determination R for the validation process is also equal to (1), indicating that the target explains 100% of the variations in the network's outputs. As for the coefficient of determination R for the testing process, it is also equal to (1), signifying that the target explains 100% of the variations in the network's outputs. Regarding the overall coefficient of determination R the network as a whole, it's equal to (100%), indicating that the target explains 100% of the variations occurring in the network's outputs. This underscores the strength of the model in predicting the quantities of carbon dioxide emissions in Iraq.

5) Step 5: Forecasting. The last step is the prediction or forecasting phase based on the inputs, which comes after building a neural network model and verifying its accuracy. The training set, validation set, and testing set are the three sets of data that are separated out. Randomly selected cases are to be predicted using the testing set. The following neural network model will be used to predict the response variables after all features have been entered into the model.

$$\hat{y}_j = f\left(\sum_{i=1}^{n} x_i w_{ij} + b_j\right) \tag{8}$$

The prediction results are presented in Table III.

TABLE IIIII
FORECAST VALUES CO_2 (AVERAGE PER CAPITA ALLOCATION IN METRIC TONS

Years	Forecast values (Average per capita allocation in metric tons.)
2022	3.649
2023	3.679
2024	4.045
2025	4.194
2026	3.895
2027	3.743
2028	3.885

After that, the performance of artificial neural networks (ANNs) in forecasting were that mean Squared Error (MSE = 0.1325) and root mean squared error (RMSE= 0.3641) are low which are good indicators of small forecasting errors. The coefficient of determination (R^2 = 0.946) is quite high; therefore, it reflects the fact that the model can explain 94.6 % of variance of the actual data quite well. The mean absolute percentage error is only 8.01%, which is low and implies that less than 10% can be a symbol of excellence. Therefore, the results demonstrate a very good forecasting efficiency by the neural network.

According to the model's conclusions, Iraq's carbon dioxide emissions should increase gradually over the coming years. In 2028, it is anticipated that per capita emissions will average 3.885 metric tons. This increase could be caused by a variety of things, such as changes in infrastructure, population growth, and economic growth. Remember that these findings are just approximations and therefore, the actual amounts of carbon dioxide released might be different.

Iraq's carbon dioxide emissions can be reduced through various methods. One possibility would be to decrease Iraq's reliance on fossil fuels by investing in renewable energy sources. Another approach would be to introduce environmental protection laws geared towards the industrial, commercial, and transportation sectors. The government of Iraq should incorporate these considerations into its future policies. By reducing its carbon dioxide emissions, Iraq can play a part in the global effort to combat climate change. Fig.

10 shows the forecast for CO_2 emissions in Iraq until the next year 2030.

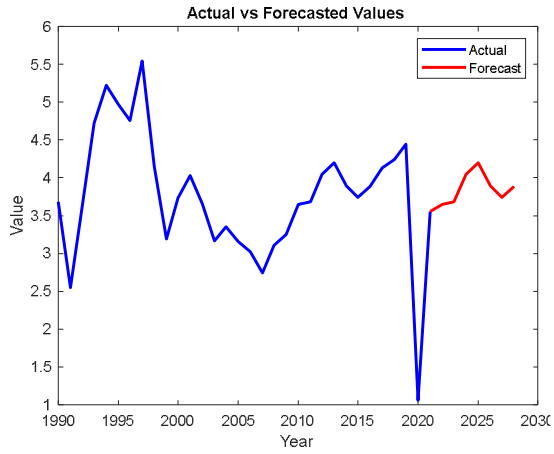


Fig. 10 Actual and forecasted values of CO_2

IV. Conclusions

Carbon dioxide emissions in Iraq are expected to decrease at a moderate or slow pace over the next decade. As shown in the results, the per capita emissions of landfills will be 3.885 metric tons by 2028, as forecasted by ANN results. This occurs due to population growth and infrastructure and economic development, which play leading roles in the increase. On the other hand, the implementation of environmental protection laws and the establishment of alternatives not related to natural gas, such as renewable energies for transport, may lead to a reduction in carbon dioxide emission rates in Iraq. Therefore, the Iraqi government needs to consider these factors while developing future policies. Mitigating the release of carbon dioxide enables Iraq to contribute to the fight against global warming.

The accuracy of predicting carbon dioxide emissions in Iraq is a realization of the fact that we no longer need to depend solely on results for proof. Neural networks use historical emission data and influencing factors to learn non-linear patterns for predicting future changes. Moreover, the network models employ methods like regularization and overfitting control to optimize performance, making the consequences more reliable.

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